



DEEP LEARNING PROJECT

CIFAR-10 Image Classification

From a shallow CNN baseline to residual networks and transfer learning

Team

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What we'll cover



The problem

Why CIFAR-10 is a meaningful benchmark



The data

60k images, 10 classes, augmentation strategies



Architectures

Shallow, Deep, Residual and Transfer Learning families



Hyperparameters

Does targeted hyperparameter tuning improve model performance?



Results

Quantitative and qualitative analysis across all models



Conclusions

Answers to our five research questions

Classifying small, noisy images into 10 categories

CIFAR-10 is a long-standing computer vision benchmark: 60,000 32×32 colour images split across 10 everyday object classes. Despite their tiny size, the images contain enough visual signal to separate similar categories — but also enough ambiguity (cat vs. dog, deer vs. horse) to make the task a genuine test of representation learning.

Goal

Build and compare multiple CNN architectures of increasing complexity — and explore targeted hyperparameter choices — to evaluate what most improves generalization on this benchmark.

Research Questions

- RQ1 — Can a custom CNN classify CIFAR-10 effectively?
- RQ2 — Does targeted hyperparameter tuning improve model performance?
- RQ3 — Does data augmentation reduce overfitting?
- RQ4 — Which classes are most often confused?
- RQ5 — Does transfer learning beat custom CNNs?

CIFAR-10 at a glance

60,000

total RGB images

10

balanced classes

32×32

pixels per image

**45k / 5k /
10k**

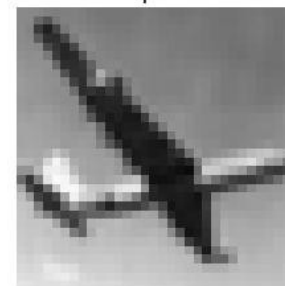
train / val / test split

Sample images from CIFAR-10

horse



airplane



truck



ship



bird



cat



automobile



ship



cat



Preprocessing and data augmentation

Pixel values are rescaled and per-channel normalized inside the model graph (Keras Rescaling + Normalization layers), so the same preprocessing applies at both train and inference time.

Two augmentation regimes were compared across multiple architectures:

Standard augmentation

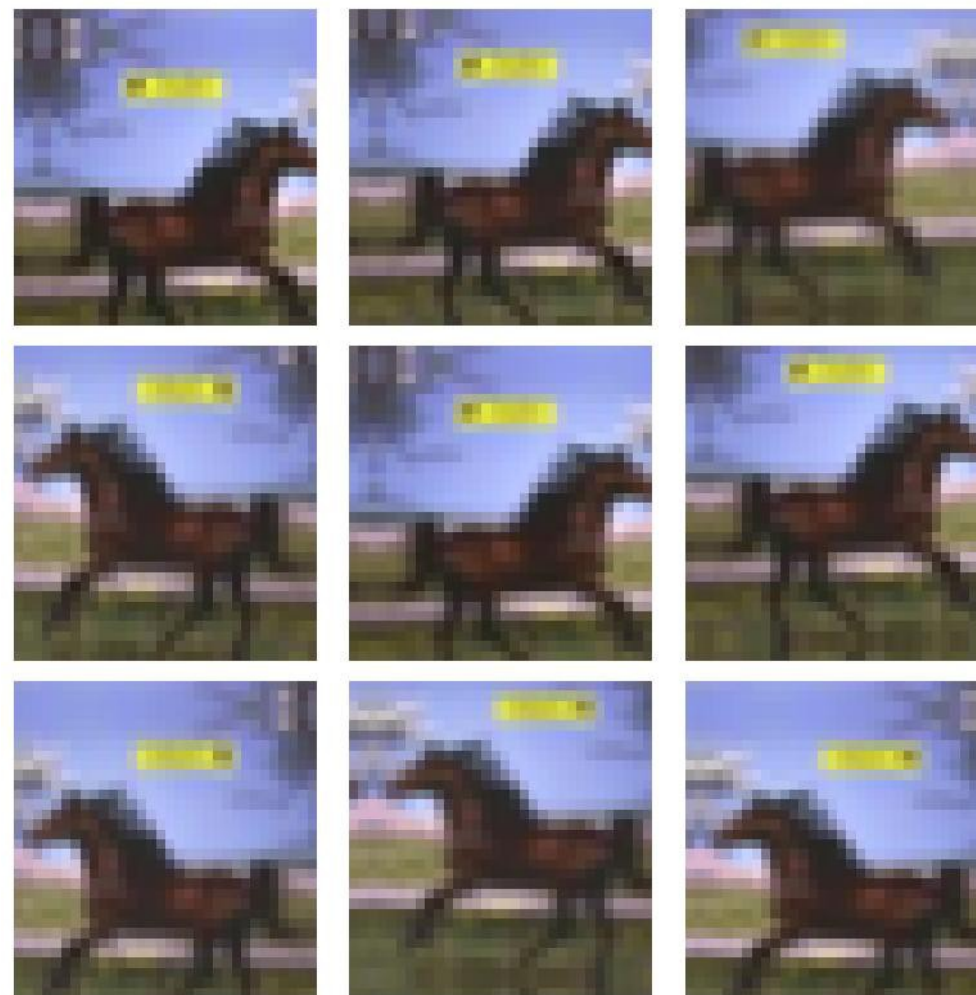
- Random horizontal flip
- Small translation
- Small contrast variation

Stronger augmentation

- Random horizontal flip
- Wider translation
- Random zoom
- Stronger contrast variation

⚠ Stronger augmentation often hurt accuracy — too much distortion at 32×32 resolution.

Augmentation preview: same image, 9 random variants



Same image, 9 augmented variants

Four architecture families, multiple variants



15 models trained in total. All share the same 45k/5k/10k split and the same evaluation metrics — differences reflect design and hyperparameter choices, not pipeline differences.



Best result: Residual CNN — 91.6% test accuracy

Shallow CNN: baseline and augmented variant

Two conv + max-pool blocks (32 → 64 filters), Dense(256) head with Dropout.
~1.07M parameters. Same architecture trained twice: without augmentation (Model 1) and with standard augmentation (Model 1b).

72.8%

Model 1 — test acc

79.7%

Model 1b + Aug — test acc

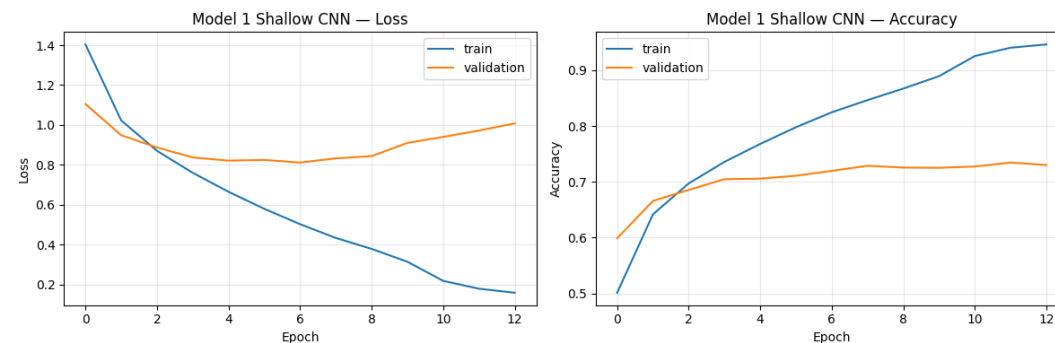
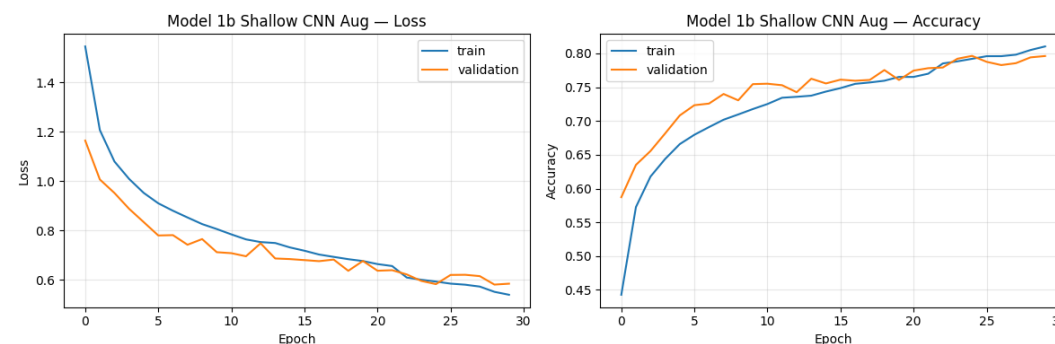
0.820

Model 1 — test loss

0.602

Model 1b — test loss

Augmentation alone: +6.9 pts accuracy, train/val gap closed → direct answer to RQ3

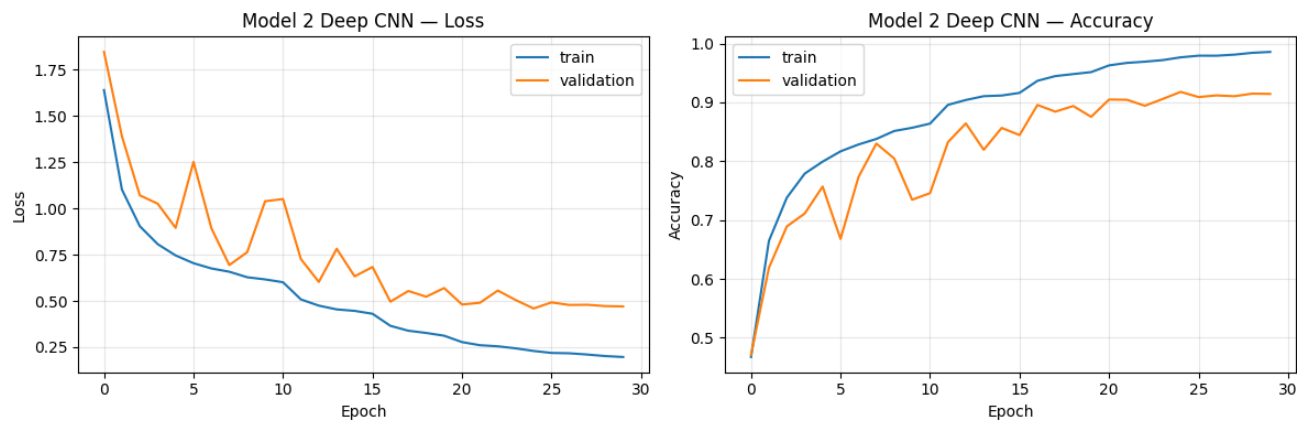
*Model 1 — Shallow CNN (baseline)**Model 1b — Shallow CNN + Augmentation*

Deep CNN: architecture and targeted hyperparameter tuning

Eight conv layers in four blocks (64→128→256→256 filters) with BatchNorm, GlobalAveragePooling and a dropout head (~2.47M params). Five training variants explored:

Variant	Test Acc	Test Loss	Observation
Deep CNN (lr=1e-3)	91.2%	0.466	Scheduler-driven LR decay — best Deep CNN variant
Deep CNN (lr=5e-4)	89.5%	0.472	Lower initial LR, same callbacks - slightly worse
+ clipnorm	89.9%	0.488	Gradient clipping adds stability, minor accuracy cost
+ L2 2e-4	89.2%	0.535	Extra L2 regularization over-constrains the model
+ Dense 256	90.2%	0.466	Smaller 256-unit head; competitive result

Deep CNN: training dynamics and per-class results



Training & validation — Deep CNN $lr=1e-3$

91.2%

test accuracy

0.466

test loss

2.47M

parameters

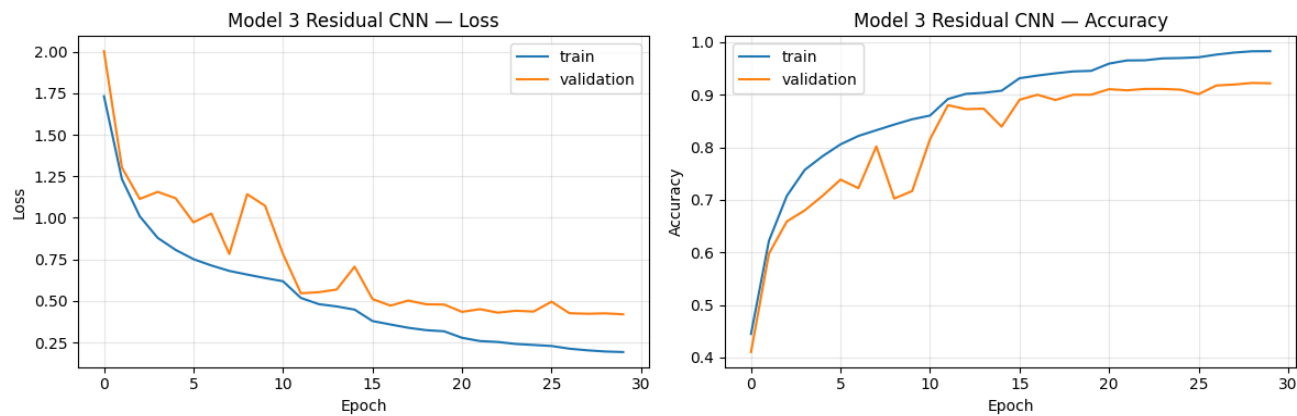
+18.4 pts over shallow baseline — direct answer to RQ2

Per-class F1 score — Deep CNN

airplane	0.920	automobile	0.963
bird	0.885	cat	0.825
deer	0.907	dog	0.863
frog	0.920	horse	0.937
ship	0.950	truck	0.949

Weakest: cat (0.825) and dog (0.863)

Residual CNN: training dynamics and per-class results



Training & validation — Residual CNN $lr=1e-3$

91.6%

test accuracy — best

0.420

test loss

2.78M

parameters

+0.4 pts over Deep CNN — skip connections ease gradient flow

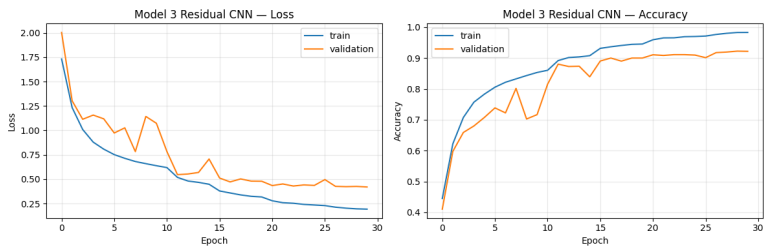
Per-class F1 score — Residual CNN

airplane	0.927	automobile	0.960
bird	0.877	cat	0.837
deer	0.916	dog	0.866
frog	0.931	horse	0.952
ship	0.951	truck	0.943

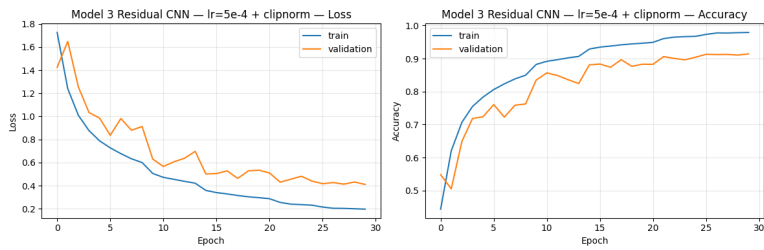
Best model overall — all classes above 0.837

Residual CNN: ablation variants

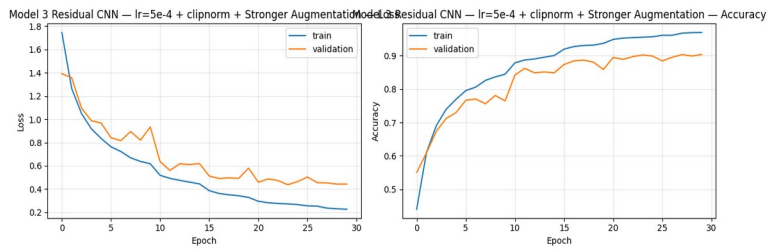
Variant	Test Acc	Test Loss	Key finding
Residual CNN (lr=1e-3)	91.6%	0.420	Best overall — default scheduler + standard aug
+ lr=5e-4 + clipnorm	91.2%	0.419	Stable training, -0.4 pts; clips gradient norm to 1.0
+ Stronger Augmentation	90.2%	0.453	Wider transforms destroy fine detail at 32×32



lr=1e-3 (91.6%, loss 0.420)



lr=5e-4 + clipnorm (91.2%, loss 0.419)



Stronger Aug (90.2%, loss 0.453)

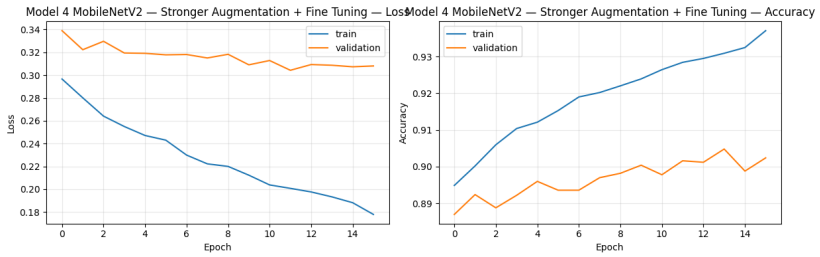
Stronger augmentation likely removes useful fine details at 32×32.

Transfer learning: MobileNetV2 and ResNet50

ImageNet-pretrained backbones (frozen) with custom heads. Five variants tested — fine-tuning, stronger augmentation and backbone size:

Model	Acc	Loss	Observation
MobileNetV2 (frozen)	88.4%	0.341	Competitive frozen baseline
MobileNetV2 Fine-tuned	89.0%	0.365	Partial unfreeze: +0.6 pts
MobileNetV2 + Stronger Aug	88.3%	0.349	Stronger aug hurts here too
MobileNetV2 Aug + Fine-tuned	90.0%	0.320	Best transfer learning result
ResNet50 (frozen)	87.7%	0.371	Larger backbone — head undertrained

MobileNetV2 Aug + Fine-tuning



Training & validation — fine-tuning phase

90.0%

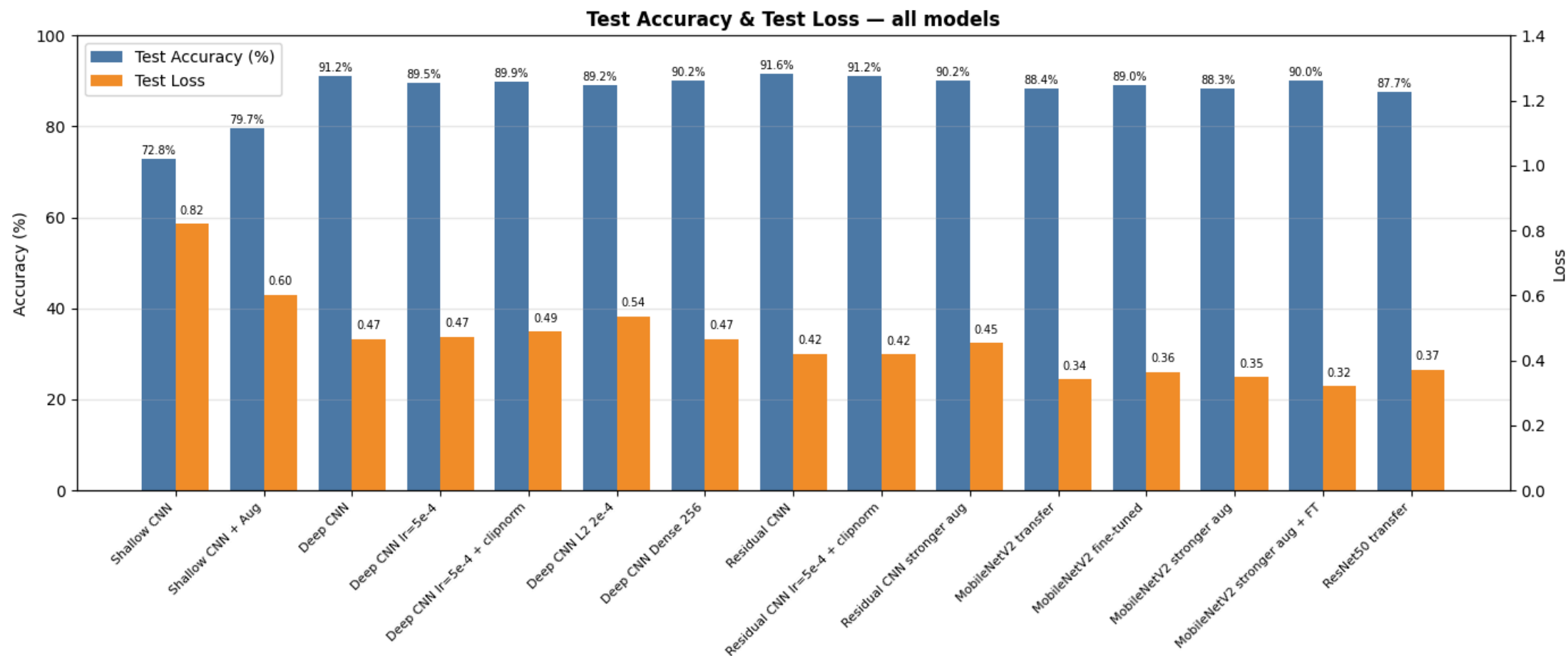
test accuracy

0.320

test loss

Best transfer learning result

Test accuracy & loss — all models

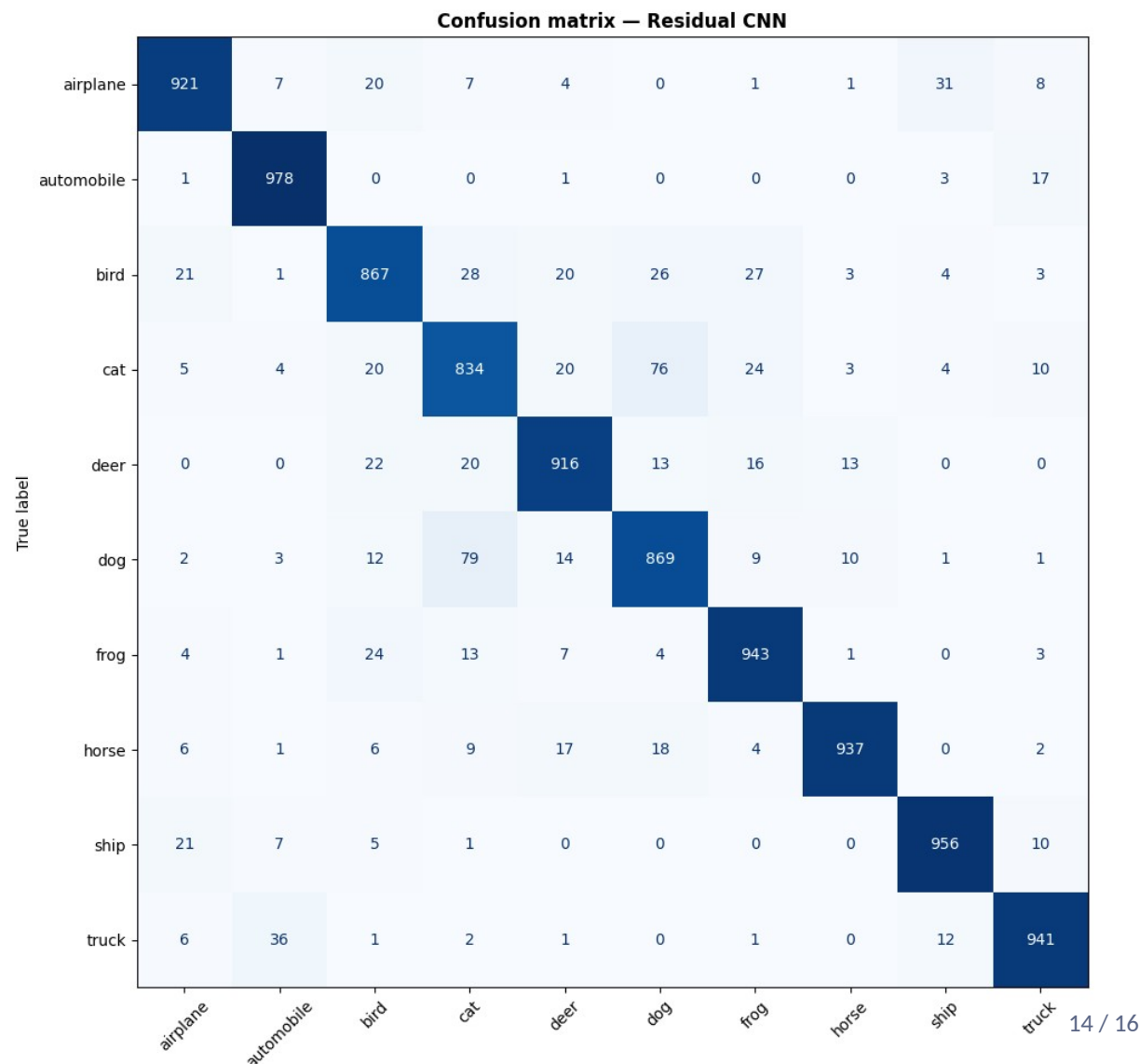


Confusion matrix: Residual CNN (91.6%)

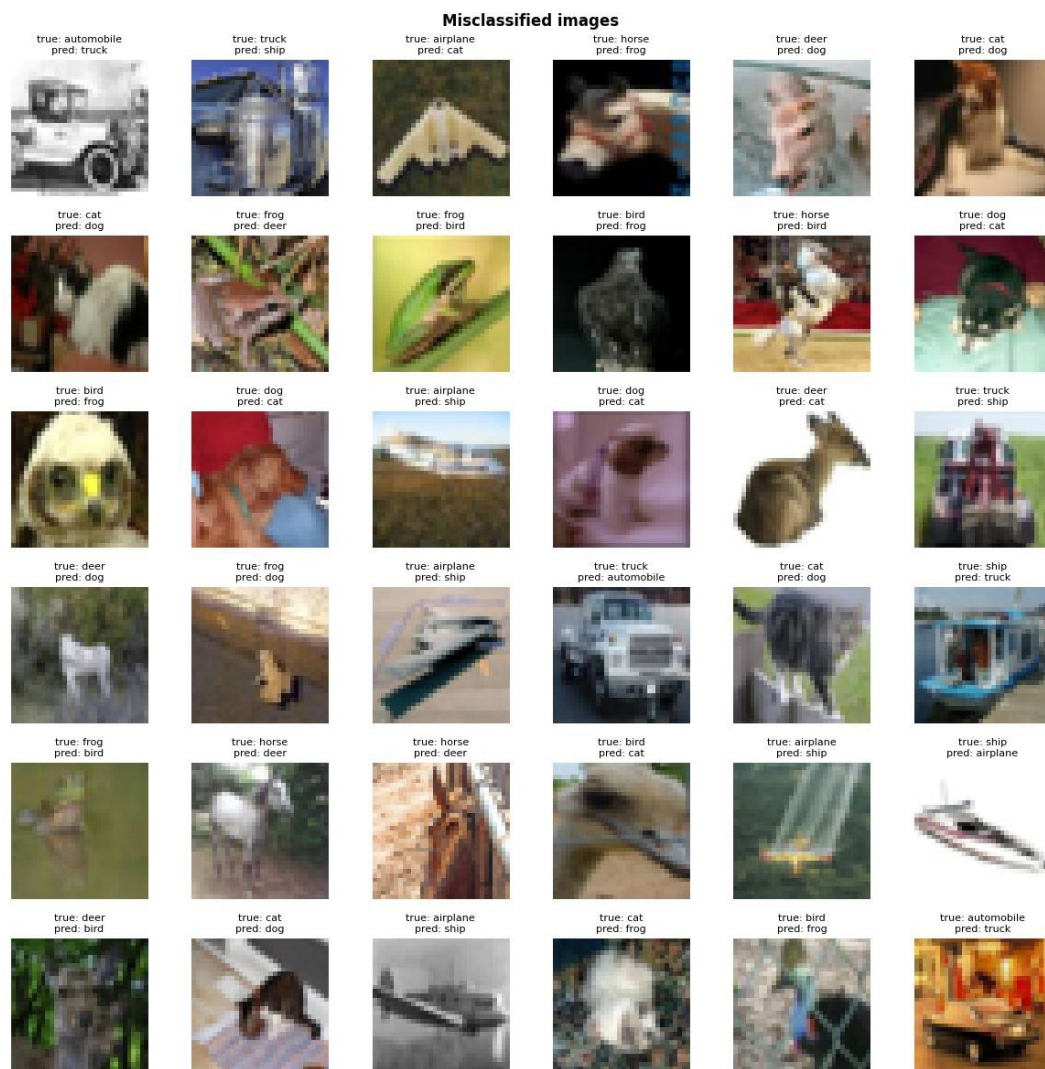
Using the best model, errors concentrate on visually similar pairs:

- cat ↔ dog — 76 + 79 confusions, largest pair
- bird ↔ frog / cat — small textured animals
- deer ↔ horse — similar four-legged silhouettes
- automobile ↔ truck — both wheeled vehicles
- ship ↔ airplane — large objects, similar colour

Residual CNN — test accuracy 91.6% · test loss 0.420



Misclassified images and what they reveal



CIFAR-10 Image Classification

What the errors look like

Most misclassified images are genuinely ambiguous at 32×32: low light, unusual poses, occlusion or very low contrast. Several errors are defensible even to a human viewer.

- cat ↔ dog and deer ↔ horse are the most frequent confusion pairs — visually similar at this resolution.
- Textured animals (bird, frog) are often confused with each other and with cats.
- Vehicles (automobile ↔ truck, airplane ↔ ship) share shape and colour cues.
- Low-light and partial-occlusion images are hard even for humans — these errors are expected.
- The best model (Residual CNN, 91.6%) still misclassifies ~840 out of 10,000 test images.

CONCLUSIONS

Answering our research questions

RQ1

Yes — a simple CNN reaches 72.8%. Standard augmentation alone lifts it to 79.7%.

RQ2

Partially — targeted hyperparameter changes affected stability and loss, but did not consistently improve accuracy. The default $\text{lr}=1\text{e-}3$ setup with ReduceLROnPlateau remained the strongest configuration.

RQ3

Yes — standard augmentation reduces overfitting. Stronger augmentation alone often hurts at 32×32 , although it helped MobileNetV2 when combined with fine-tuning.

RQ4

Cat/dog, deer/horse and automobile/truck are the dominant confusions — all visually similar pairs.

RQ5

Partially — MobileNetV2 Aug + Fine-tuned reaches 90.0%, ResNet50 87.7%; both trail the best custom CNN (91.6%).

